

The Sound of Water: Inferring Physical Properties From Pouring Liquids

Piyush Bagad , Makarand Tapaswi , Cees G. M. Snoek, and Andrew Zisserman 

Abstract—We study the connection between audio observations and the underlying physics of a mundane yet intriguing everyday activity: *pouring liquids*. Given only the sound of liquid pouring into a container, our objective is to automatically infer physical properties such as the liquid level, the shape and size of the container, the pouring rate and the time to fill. To this end, we: (i) show in theory that these properties can be determined from the fundamental frequency (pitch); (ii) train a pitch detection model with supervision from simulated data and visual data with a physics-inspired objective; (iii) introduce a new large dataset of real pouring videos for a systematic study; (iv) show that the trained model can indeed infer these physical properties for real data; and finally, (v) we demonstrate strong generalization to various container shapes, other datasets, and in-the-wild YouTube videos. Our work presents a keen understanding of a narrow yet rich problem at the intersection of acoustics, physics, and learning. It opens up applications to enhance multisensory perception in robotic pouring.

Index Terms—Audio-visual learning, physics-based learning, liquid pouring.

I. INTRODUCTION

WHAT can possibly be scientifically interesting about such a mundane chore as pouring a liquid into a glass? We perform this action all the time but barely realise that we effortlessly learn to infer several useful physical properties in the process. For example, evidence in psychoacoustics suggests that humans can accurately infer the level of the liquid, the time to fill [1], the size of the container [2], and even the temperature of the liquid [3], [4], [5], merely from the sound of pouring. Such inference (e.g., time to fill) allows us to adaptively control our actions (e.g., stopping pouring to prevent spillage) conforming to the *affordance* theory by Gibson [6]. In this work, we study the physical phenomenon involved in liquid pouring and explore how it can be used to train machines to infer useful physical properties from sound alone.

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Despite its mundaneness, liquid pouring has rich physics underpinning it and has been studied for more than a century [7]. The crux of this exploration is summarized well by Berg and Stork [8]: “As the liquid (e.g., water) is filled, a sound consisting of an increasing pitch and some (odd) harmonics superimposed with whooshing, gurgling is observed”. This pitch and the corresponding harmonics are a function of the physical properties. For example, the shape of the pitch depends on the container shape [9], [10], [11], the range of the pitch depends on the container dimensions [1], and the rate change of pitch depends on the pouring flow rate [1]. Thus, automatically inferring useful physical properties from the sound of pouring necessitates two stages: (i) detecting the pitch from the raw audio signal, and (ii) recovering these physical properties from the pitch. There are several challenges in training machines to do these purely from sound.

First, such a task requires a fine-grained, time-sensitive understanding of audio while contemporary models focus more on coarse recognition tasks like sound classification [12], [13], [14]. Second, the underlying physics of such a niche activity as pouring is not fully developed for a general container and liquid setup, unlike, say, Newtonian mechanics studied analogously in [15]. Third, there is a lack of clean, controlled, and large datasets which are necessary to study physical property estimation by learning. Fourth, supervision either in the form of pitch annotation or the actual physical properties is difficult to obtain and to use directly in training.

To enable a systematic study, we collect a clean and large dataset of 805 videos of pouring across 50 diverse containers. For training, we select a subset of containers shaped like cylinders such that we can approximate the underlying physics with that of a cylinder [1]. We design an audio network for pitch detection based on wav2vec2 [16] pre-trained on speech data that has characteristic pitch dynamics [17]. As pitch annotations are hard and ambiguous to obtain at scale, we use supervision from simulated data and visual data. We pre-train the network on simulated sounds of liquid pouring. On real data, we fine-tune the network by visual co-supervision with a physics-inspired objective. We demonstrate that the co-supervised audio model is able to predict pitch, and hence estimate physical properties, with a performance far exceeding that of multiple previous methods. Note that visual observations are only used as part of the training to obtain pseudo labels and only audio is used during inference. An overview of the model is shown in Fig. 1.

Why is this important, though? First, as far as we know, this is the first work to demonstrate human-like capabilities (or better)

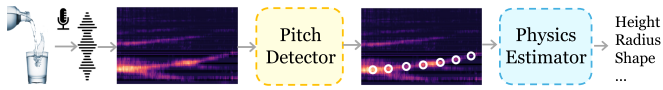


Fig. 1. Overview: An overview of the objective and method. We train a pitch detector without any manual supervision and rely on physics to estimate physical properties merely from the sound of water.

in predicting physical properties from sound alone – in fact, we achieve an accuracy of ± 0.60 cm in predicting the air column height for cylinders. Second, although the model is trained on cylinders, we show that the pitch estimation (and the model in general) is applicable beyond cylinders – for example, it can be used to predict the shape of containers with convincing accuracy. Third, the model generalizes well to videos from other datasets and to in-the-wild YouTube videos. In summary, our contributions are:

- 1) We show in theory that the physical properties of the container-liquid system can be recovered from the fundamental frequency (pitch) of the sound of pouring.
- 2) We train a pitch detection model using supervision from simulated data and visual data with a physics-inspired objective.
- 3) We introduce a new clean and large dataset of videos of liquid pouring that can be used to study the estimation of physical properties.
- 4) We demonstrate that the model is indeed capable of detecting pitch and hence estimates physical properties only from the sound of pouring. We show that the model generalizes to different container shapes, to other datasets, and to in-the-wild YouTube videos.

A shorter version of this work appeared in ICASSP 2025 [18].

II. RELATED WORK

a) Psychoacoustics of pouring: Psychoacousticians have demonstrated a remarkable human ability to infer physical properties such as material [19], [20], shape [21], and size [22], [23], [24] from sound alone. Furthermore, the temporal evolution of sound also provides an anticipatory sense in humans that helps to estimate dynamic properties such as distance [25], velocity [26], and time-to-contact [27]. Interestingly, liquid pouring presents a unique case in which humans can infer both static properties (e.g., container size, material, liquid temperature, etc.) and dynamic properties (e.g., liquid level, time-to-fill) from sound alone [1], [2], [3]. We draw inspiration from this line of work and train audio models to infer physical properties from the sound of pouring.

b) Pouring in the literature: Pouring occurs surprisingly often in the literature. In the machine learning community, pouring has been studied by roboticists to learn how to pour [28], [29], [30], [31], [32], [33], [34], [35], [36]. In computer vision, there has been much work on visually perceiving liquids either in a static setting [37], [38], [39], [40], or during pouring [41], [42], [43], [44], [45]. For example, [37], [38], [41] detect the amount of liquid in a container, [39] detect the container shape and material, and [42], [44] track the stream of pouring liquid. Likewise,

there has been work on estimating the dynamic latent states (e.g., height or mass of liquid at a given time) from multi-modal (vision, audition, haptics, etc.) inputs [46], [47], [48], [49], [50], [51]. The majority of these works [48], [49], [50], [51] use additional sensory data (e.g., force, torque, hand trajectory, inertial measurements) in combination with vision or audio or both. Such measurements require sophisticated recording equipment. In contrast, our aim is to be able to predict physical properties from sound alone, and to achieve this without using bespoke equipment, but instead from regular smartphone recordings of liquid pouring. Closest to our work is that of Wilson et al. [46] where an audio-visual CNN is supervised to predict the mass of liquid poured at a given time, given instantaneous video and audio clips. Similarly, Wang et al. [52] also predict fill level from audio-video inputs. Methodologically, our work differs from [46] and [52] by incorporating the underlying physics directly in the learning process. By design, our method can estimate several physical properties and not only the liquid mass without supervision. We also evaluate our model by linear probing of the co-supervised features on the dataset of [46] and report superior performance.

c) Liquid and pouring datasets: Most robotic pouring work uses private datasets recorded with a robotic arm in a lab setting, or simulated data to train robots. Although [53], [54], [55] records third-person pouring videos, these datasets are either not openly available, very small in scale, or have missing audio recordings. Pouring does feature in popular large-scale audio-visual datasets such as VGGSound [13], AudioSet [14] and EPIC-Kitchens [56]. But these are very limited in number and too visually noisy (large viewpoint changes, low visibility, occlusions) for a controlled study. Likewise, there are lots of pouring videos on YouTube (Shorts in particular), but these too are visually noisy. Closest to our requirements is the dataset collected by Wilson et al. [46] of 500+ videos, but only 276 of them have liquid pouring across only 4 containers. So, we resort to recording our own dataset of 805 videos of pouring across 50 containers with a casual smartphone camera in a domestic setting. We have publicly released our dataset to stimulate further research.¹ In addition, we also evaluate our models on the dataset proposed by Wilson et al. [46].

d) Audio-visual learning: The natural audio-visual correspondence in videos coupled with large-scale video datasets [13], [14] has led to an array of work on self-supervised representation learning for various downstream tasks [57], [58], [59], [60], [61], [62]. These approaches can be broadly categorized as contrastive [58], [63], [64], generative [60], [61], [65], paired sample discriminative [57], [66], [67], [68], [69], clustering [59], [62], [70] and distillation-based [71], [72]. More recently, with the proliferation of transformer-based language models, audio representations have been learned together with text and video [64], [73], [74], [75], [76]. Beyond learning general representations, there has also been much work on special downstream tasks such as audio visual localization [68], [77], [78], lip reading [79], [80], sound/visual generation [81], [82], [83]. These approaches and tasks largely ignore the fine-grained time-dependent changes in

¹<https://huggingface.co/datasets/bpiyush/sound-of-water>

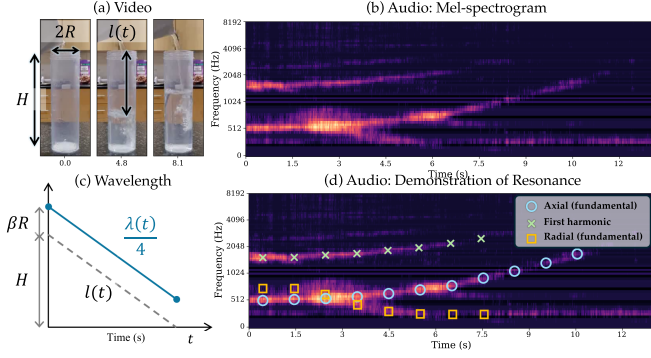


Fig. 2. Demonstration of resonance in liquid pouring. As liquid is poured in the container shown in (a) of height H and radius R , a sound made up of an increasing pitch (fundamental frequency) and some (odd) harmonics is observed on the spectrogram shown in (b). Two kinds of resonance are observed: axial (fundamental shown as blue circles in (d), first harmonic as green crosses) and radial (fundamental shown as yellow squares). The wavelength (inverse of frequency) of the axial resonance (shown in (c)) is a function of the length of air column $l(t)$: $\lambda(t)/4 = l(t) + \beta R$. Interestingly, the high intensity blob around 3 s is likely due to the mixture of pitch from both kinds of resonance.

sounds and rely on instantaneous and coarse correspondences. In contrast, liquid pouring necessitates modeling of fine-grained characteristics over time (namely, pitch). Most recently, the prominence of LLMs has spurred research on integrating audio in language models [84]. Likewise, video generation models such as Veo 3 [85] and others now include realistic audio outputs. While these are extremely competent models, it is unclear if such models can be used to decode such fine-grained physical properties from sound alone.

III. THE PHYSICS OF LIQUID POURING

As an example case, consider a simple cylindrical vessel of radius R , height H as shown in Fig. 2(a). At time t , suppose that the vessel is filled to a level such that the length of the air column is $l(t)$. While water is poured, we hear a mix of pitch and odd harmonics that correlate with the length of the air column at a given time. This is visible on the spectrogram in Fig. 2(b). We term this resonance as *axial* resonance. The observed pitch is shown as the blue curve on the spectrogram in Fig. 2(d). The corresponding wavelength that varies linearly w.r.t. $l(t)$ is shown in Fig. 2(c). Here, we describe the physical equation that determines this curve and the physical properties derived from it. We also discuss another kind of resonance, termed *radial* resonance, which is less prominent but co-occurs with axial resonance.

A. Axial Resonance

As the water fills up, it pushes out air in the air column creating a frequency pattern that resembles blowing air in an organ pipe closed at one end. This phenomenon has been studied by physicists for a long time [9]. As the water level increases, the vacant space for air molecules to vibrate reduces and hence the frequency increases. This frequency can be described mathematically with an analogy to a string tied at one end. At time t ,

the fundamental frequency $f(t)$ is given by

$$f(t) = \frac{c}{4} \frac{1}{l(t)}, \quad (1)$$

where c is the speed of sound in air. This expression arises from a standing wave of wavelength $\lambda(t) = 4l(t)$ where the amplitude is zero at the water surface and maximum at the top of the vessel. Rayleigh [7] and others studied this and found an experimental end-correction that depends on the radius of the container:

$$f(t) = \frac{c}{4} \frac{1}{(l(t) + \beta R)}, \quad (2)$$

where β is an end-correction factor generally agreed to be 0.62 [86]. The numerical value of β has also been debated in the acoustic-physics community [86], [87], [88] but we fix it to 0.62. A spectrogram of the sound of pouring in a sample container is shown in Fig. 2(b). The observed pitch $f(t)$ (blue circles) and first harmonic (green crosses) are marked on the spectrogram in Fig. 2(d).

To avoid working with an inverse relation, we look at this equation in terms of wavelength $\lambda(t)$,

$$\lambda(t) = \frac{c}{f(t)} = 4(l(t) + \beta R). \quad (3)$$

Note that all these quantities are in metric units. The LHS is observable from the audio of liquid pouring while the RHS is observable from a video of liquid pouring (up to a scale factor). Fascinatingly, this implies that the audio is effectively a *metric ruler* for objects in the video.

B. Radial Resonance

In addition to axial resonance, another kind of resonance is observed when water is poured into a vessel. This is subtle and not directly visible or even clearly audible, but it does show up when carefully analyzing spectrograms. As water is poured, it generates vibrations on the surface of the container and the container vibrates radially. As the liquid level increases, the mass of the combined container-liquid system increases, which acts as an inertia that decreases the vibration frequency. Thus, in this case, as the liquid level increases, the frequency decreases. This has been studied in [9], [11] and is also related to the notion of musical glasses [89]. Formally, this frequency for cylindrical containers is given by [9]:

$$f(t) = \frac{f_0}{\left[1 + \xi \left(1 - \frac{l(t)}{H}\right)^3\right]^{1/2}}, \quad (4)$$

where f_0 depends on the container's dimensions and physical properties such as density, thickness, and ξ depends on the liquid density as well as container material density. In this case, the observed frequency curve is shown with yellow squares in Fig. 2(d).

C. Recovering Physical Properties From Pitch

Our main observation is that to determine some of the key physical properties, it suffices to estimate the *fundamental wavelength* in **axial** resonance from raw audio. Recall that the basic

physical relation driving this system is given by (3). We categorize the physical properties in two sets. (i) *Static properties*: these are inherent to the container-liquid system (e.g., container size) and do not vary over time. (ii) *Dynamic properties*: these are a function of time (e.g., length of the air column, flow rate, time to fill). We start with the derivation for the length of air column and then derive other properties from that.

- i) *Length of air column*: We want to estimate $l(t)$ given $\lambda(t)$ at a given time t . If we also know l, λ at another point $t' \neq t$, then we get:

$$l(t) = l(t') + \frac{1}{4} [\lambda(t) - \lambda(t')].$$

Using the boundary condition $l(T) = 0$, where T is the total pouring duration, we get:

$$l(t) = \frac{1}{4} [\lambda(t) - \lambda(T)]. \quad (5)$$

- ii) *Container size*: Container height and radius are directly obtained from the boundary conditions:

$$H = l(0) = \frac{\lambda(0) - \lambda(T)}{4} \quad \text{and} \quad R = \frac{\lambda(T)}{4\beta}. \quad (6)$$

- iii) *Volume flow rate*: Likewise, we can derive the other properties. For the volume flow rate $Q(t)$, suppose the volume at time t is $V(t)$. Then,

$$Q(t) = \frac{dV}{dt} = -\pi R^2 \frac{dl}{dt} = -\frac{1}{4} \pi R^2 \frac{d\lambda}{dt}, \quad (7)$$

where the derivative $\frac{d\lambda}{dt}$ can be approximated using the estimated $\lambda(t)$.

- iv) *Time to fill*: For time to fill, we assume a constant flow rate (since otherwise, one could pause pouring midway leading to ill-defined time to fill). Also, we do not know the true duration T and are only given a partial audio, i.e., cut upto time t . Here, following Cabe et al. [1], we make an additional assumption that the end-correction term βR is small at the start of pouring ($\beta R \ll H$). Thus, in a short interval at the start of pouring $t' \in (0, \delta)$, ignoring the end correction, we can approximate

$$\tau(t') = - \left[\frac{l(t')}{\frac{dl}{dt}} \right] \approx - \frac{\lambda(t')}{\frac{d\lambda}{dt}}, \quad \forall t' \in (0, \delta). \quad (8)$$

Then, we can use the property of τ to get time to fill at any given time t :

$$\tau(t) = T - t = (T - t') + t' - t = \tau(t') - (t - t'),$$

for some $t' \in (0, \delta)$, $\delta \ll t$. Note that this needs reliable estimates of λ and its derivative at the beginning of the audio.

Hence, we have shown that to estimate the desired physical properties, it suffices to determine the fundamental wavelength $\lambda(t)$, $\forall t$. Note that we require precise estimation of $\lambda(t)$, $\forall t \in [0, T]$ and particularly at the start ($t = 0$) and end ($t = T$).

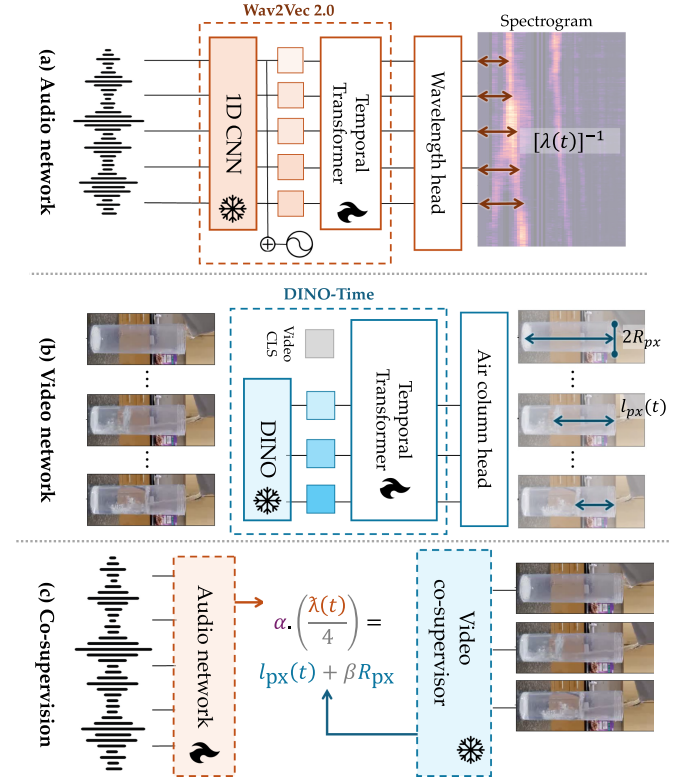


Fig. 3. Model architecture and training. (a) The **audio network** is based on a wav2vec2 repurposed for pitch detection. (b) The **video network** is based on DINO repurposed to operate on image sequences to detect length of air column and container radius (up to a scale factor). (c) The audio network is pre-trained on synthetic samples and then fine-tuned on real samples using physics-inspired co-supervision from the video.

IV. AUDIO NETWORK AND TRAINING

Our objective is to predict physical properties (e.g., length of air column, size of the container) from the sound of pouring. Our approach is formulated as a two-stage process: (i) detecting the pitch from the raw audio signal, and (ii) recovering these physical properties from the pitch. To detect pitch, we propose a network based on wav2vec2 (Section IV-A). We pre-train it on simulated sounds of liquid pouring (Section IV-B). Then, on real data, we use the visual stream to co-supervise the audio network (Section IV-C). Given a strong pitch detector, we use (5), (6), (7), and (8) to obtain the desired physical properties from the pitch.

A. Audio Network Architecture

The network takes in raw audio samples and outputs wavelength (pitch) estimates at each time step. The architecture is based on wav2vec2 [16] adapted for pitch detection on pouring sounds. The architecture diagram is shown in Fig. 3(a). The network takes in raw audio waveform and outputs a distribution over the set of wavelength bins at each time step. Note that we predict the fundamental wavelength as opposed to fundamental frequency because the wavelength varies linearly with the length of the air column and we want to bake in this linearity in the

learned features. The input waveform is resampled at a rate of 16 k Hz. First, the waveform is tokenized using a 1D CNN encoder which takes in windows of 400 samples (25 ms) of audio with a hop length of 320 samples (20 ms). In addition to the original model design, we add sinusoidal position embeddings to the tokens to enhance temporal information. These are then passed through a Transformer network with 12 blocks (model dimension 768, 8 attention heads). This is followed by a prediction head, a linear regressor that maps from $\mathbb{R}^{768} \rightarrow \mathbb{R}^K$ where K is the number of wavelength bins. The output is converted into a distribution over wavelength using Softmax activation. More details on the architecture are provided in the Supplemental.

a) Training the network: We want to train the described network to detect pitch. Since it is difficult to obtain pitch annotations on real samples, we first pre-train the network on synthetic samples with perfect ground truth (Section IV-B). Then, we fine-tune on a small amount of real data with video as the source of co-supervision (Section IV-C).

B. Pre-Training With Synthetic Data

First, we describe how we generate simulated pouring sounds and then, we describe the pre-training details.

a) Synthetic data generation: We train a generative model based on Differentiable Digital Signal Processing (DDSP) [90] to simulate the sounds of liquid pouring. The architecture is a supervised autoencoder model with the latent space decomposed into pitch, loudness, and a residual vector. The encoder represents the raw audio waveform in three parts: (i) pitch over time, (ii) loudness over time, and (iii) residual vector. The pitch is extracted using CREPE [91], a standard pitch detector popularly used in music. Loudness is extracted as standard RMS energy. The residual is learned with a GRU network operating on Mel Frequency Cepstrum Coefficients (MFCC) features. Intuitively, the residual captures background noise and room reverberation characteristics. The decoder is composed of synthesizers based on classical signal processing techniques. It takes in pitch, loudness, and the residual and generates a realistic waveform. The network is trained using a multi-scale spectrogram reconstruction loss. To generate a sample, we first randomly pick a conditioning sample from the train set and extract loudness and residual vector from it. We can pass an arbitrary pitch profile with the chosen loudness and residual. To simulate a pitch profile, we sample dimensions of an arbitrary cylindrical container. Formally, we sample radius R , height H , and then compute length of air column as a linear curve:

$$l(t) = \left(-\frac{H}{T}\right)t + H, \quad (9)$$

where T is the duration of the conditioning sample. Then, we plug this in (3) to compute the wavelength $\lambda(t)$ which is inverted to obtain the pitch. These three (pitch, loudness, residual) are then fed to the trained decoder which generates a realistic waveform with the desired pitch profile. For a single conditioning sample, the loudness and residual are fixed while we can vary (R, H) to vary the pitch and cover a vast diversity of pitch profiles. Some examples of generated samples are shown

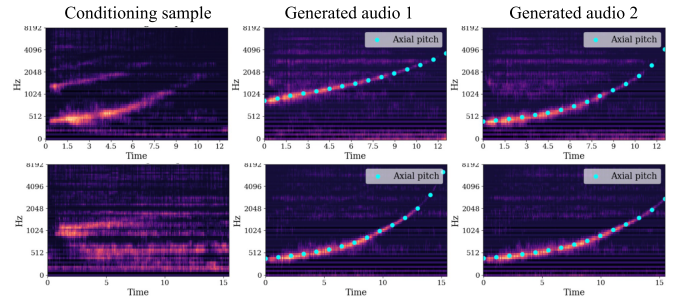


Fig. 4. Samples of simulated pouring sounds. Our simulator takes in (i) a real sample from the train set as condition, (ii) a random pitch profile and generates synthetic waveform that resembles sound of pouring liquid in a cylindrical container. More samples are shown in the Supplemental.

in Fig. 4. We sample $H \sim U[5, 25]$ cm, $R \sim U[1, 5]$ cm and generate 10,000 samples. This is randomly divided into a train and validation set.

b) Implementation details for pre-training: We train only the penultimate 8 layers of the Transformer and the prediction head keeping the rest of the network frozen. The number of wavelength bins is chosen as $K=64$. The wavelength range is chosen to be $[0, L]$, $L=100$. Each bin represents a length of $L/K=100/64=1.56$ cm. Following [91], to soften the penalty for near-correct predictions, the target is Gaussian-blurred in wavelength such that the energy surrounding a ground truth wavelength decays with a standard deviation of 1.25 bins (or 1.95 cm). The network is trained with a KL divergence penalty between the predicted and true distributions. It is trained for 100 epochs with a batch size of 32 using Adam optimizer [92] with a constant learning rate of $1e^{-4}$.

C. Fine-Tuning by Visual Co-Supervision

The pre-trained audio network needs to overcome the sim2real gap to detect pitch accurately on real samples. Since it is difficult to obtain ground-truth pitch annotations on real samples, we make use of the video stream as a source of weak supervision for fine-tuning. From the RHS in (3), the video can supply measurements for the length of the air column and the container radius (up to a scale), which we use to supervise the audio network.

a) Video pre-training with pseudo labels: To use video as a teacher for co-supervision, we pre-train a video network to detect the length of the air column (or equivalently, the level of liquid) and the average radius of the container. The architecture is shown in Fig. 3(b). We use a frozen DINO [93] encoder per frame and attach a Transformer to model the temporal dependencies. This is followed by a prediction head for the length of the air column $l(t)$ (relative to the image size). The (pseudo) labels to train this network are obtained using temporal difference between adjacent frames and classical image processing techniques (Derivative of Gaussian on temporal difference heatmaps) to obtain clean ground truths. More details are provided in the Supplemental. The radius of the container is obtained by segmenting the container from the first frame of the video using SAM [94]. Note that both these measurements are in pixel scale.

b) *Scale-aware video co-supervision*: The wavelength predictions $\lambda(t)$ from the audio network are in metric units while the length predictions $l_{px}(t)$, R_{px} from the video network are in pixels. To enable video as a weak supervisor, we need to account for a scale factor, α .

$$\alpha \cdot \left(\frac{\lambda(t)}{4} \right) = l_{px}(t) + \beta R_{px}. \quad (10)$$

This factor encapsulates the depth and camera intrinsics and is unique per video. Assuming a usual perspective camera, we can obtain a precise definition of the scale factor:

$$\alpha := \frac{1}{Z} \frac{f}{s}, \quad (11)$$

where Z is the depth, f is the focal length and s is length per pixel. Note that α is inversely proportional to depth. We can pre-compute α for each video by simply computing the ratio of wavelength from audio to the pixel lengths from video. However, since the audio estimates are generally off where the signal is low (e.g., towards the end of the pouring sequence), this leads to very poor estimates of α . To account for this, we weigh the ratios over time with the RMS energy in the audio over time. This leads to robust scale estimates. Some example containers and scale estimates are shown in the Supplemental. For example, a large container is kept further away from the camera and thus has a smaller α . Next, we fix the scale factors and the video network and fine-tune the audio network with the MSE loss to improve predictions of $\lambda(t)$ using supervision based on (10).

c) *Implementation details for fine-tuning*: We convert the audio network outputs from distributions to scalar wavelengths. The video outputs are already scalars. We use the standard MSE loss (10) and fine-tune the network for 5 epochs with the Adam optimizer [92] and a constant learning rate of $1e^{-6}$. At inference, given a sound of pouring, our model predicts a wavelength distribution which is converted to wavelength scalars and subsequently to frequency scalars. These frequencies can then be overlaid on spectrograms to verify correctness qualitatively.

V. DATASET: SOUND OF WATER 50

Our dataset consists of videos showing a human hand pouring liquid in a container with a fixed camera facing the container. The videos are recorded by the authors with a smartphone camera in a domestic setting. Across videos, we randomly vary the flow rate but keep it approximately constant within a single video. In total, we collect 805 videos across 50 containers (4 shapes, 5 materials) and 2 liquids (hot and normal water). The shapes are cylindrical, semiconical, bottleneck, and hemispherical. The materials include glass, plastics, ceramics, steel, and cardboard. Some example containers are shown in Fig. 5. Some example video sequences are shown in Fig. 6.

a) *Splits*: Since we only train with cylinder-shaped containers, we carefully create splits with a single training set and multiple test sets. Details of all four splits are provided in Table I. In short, the train set has videos of pouring in transparent cylinder-like containers. Test set I has a subset of containers of those in test (seen containers) but is distinct in terms of video sequences. Test set II and III have completely unseen containers. We use

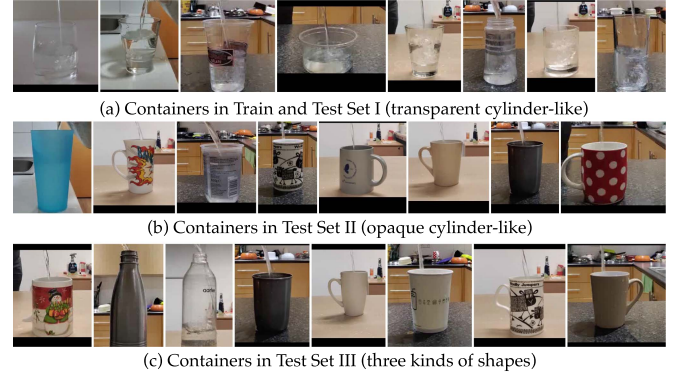


Fig. 5. Examples of containers used in the *Sound of Water 50* dataset. The dataset contains videos of pouring liquids in containers with diverse shapes, materials, opacity and background environments. The train set has videos of pouring in transparent cylinder-like containers. Test set I shares the same set of containers but is distinct in terms of the videos. Test set II and II have videos of pouring in entirely unseen containers.

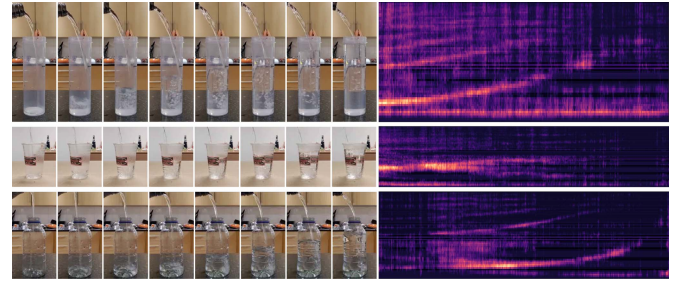


Fig. 6. Sample video sequences from the *Sound of Water 50* dataset. (Left) Sample pouring sequences in each of the three container shapes: cylindrical, semiconical and bottleneck. (Right) The corresponding spectrograms of the pouring sounds.

TABLE I

SPLITS IN *SOUND OF WATER 50* DATASET. WE CREATE FOUR SPLITS IN OUR DATASET. THE TRAIN SET CONTAINS VIDEOS OF POURING IN TRANSPARENT CYLINDER-LIKE CONTAINERS. TEST SET I CONTAINS CONTAINERS THAT ARE A SUBSET OF THOSE IN TRAIN SET BUT HAS DISTINCT VIDEO SEQUENCES. TEST SET II HAS COMPLETELY UNSEEN CYLINDER-LIKE CONTAINERS. TEST SET III ALSO HAS UNSEEN CONTAINERS INCLUDING BOTTLENECK SHAPED CONTAINERS. TEST III IS USED FOR SHAPE CLASSIFICATION AND OVERLAPS ONLY WITH TEST II IN TERMS OF CONTAINERS. THIS ADDS UP TO $18 + 25 = 43$ CONTAINERS, AND $195 + 54 + 434 = 683$ VIDEOS. THE REMAINING 122 VIDEOS (OUT OF A TOTAL OF 805) ARE OF HEMISPHERICAL/FREEFORM CONTAINERS ONLY USED FOR QUALITATIVE ANALYSIS. SOME EXAMPLE CONTAINERS IN EACH OF THESE SETS ARE SHOWN IN FIG. 5.

Split	Opacity		Shapes			Containers	Videos
	Transp.	Opaq.	Cyl.	Semi-cone	Bottle		
Train	✓	✗	✓	✓	✗	18	195
Test I	✓	✗	✓	✓	✗	13	54
Test II	✗	✓	✓	✓	✗	19	327
Test III	✓	✓	✓	✓	✓	25	434

the different test sets according to the requirements of a given task, e.g., we use *Test Set III* to probe shape classification. Some examples of containers in each split are shown in Fig. 5.

b) *Statistics*: Basic statistics from our dataset are shown in Fig. 7. Since we use a variety of container sizes, the video duration ranges from 3.5 s to 30.1 s with a mean of 10.4 s. The average

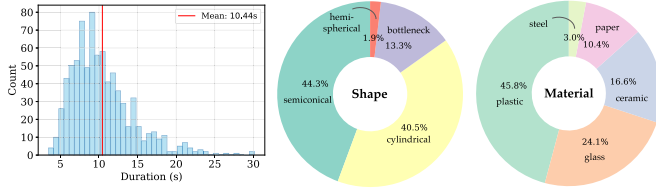


Fig. 7. *Sound of Water 50* dataset statistics. We show some basic numbers from the dataset. (Left) shows distribution of video duration, (center) shows distribution over container shapes and (right) shows distribution over container material.

TABLE II

COMPARISON WITH BASELINES IN ESTIMATING LENGTH OF AIR COLUMN. MEAN ABSOLUTE ERROR (IN CMS) IN ESTIMATING THE LENGTH OF AIR COLUMN $l(t)$ ON THE TWO TEST SETS. OUR MODELS COMFORTABLY BEAT ALL THE PITCH DETECTION BASELINES. GENERALLY, PERFORMANCE ON TEST SET II IS LOWER AS IT CONSISTS OF CONTAINERS NOT SEEN DURING TRAINING.

Method	Test set I seen containers ↓	Test set II unseen containers ↓
Baselines		
Yin [95]	30.80	27.30
PESTO [96]	11.70	10.60
CREPE [91]	7.61	9.40
argmax on spectrogram	4.60	5.11
Ours		
Audio-only	0.78	0.82
Co-supervised	0.60	0.71

height and (base) radius of a container are 11.5 cm and 3.1 cm, respectively. The containers are sourced from four shapes (cylindrical, semi-conical, bottleneck, hemispherical) and five materials (glass, plastic, steel, paper, ceramic) distributed as shown in Fig. 7.

VI. EXPERIMENTS

In this section, we present various experiments to demonstrate physical understanding from sounds of liquid pouring. We start with a brief description of our dataset. Then, we describe the results on estimating pitch (Table II) and physical properties from pitch (Table III). Then, we present results on classifying container shapes and estimating liquid weight directly from the learned representations.

A. Estimating Basic Physical Properties From Pouring Sound

As theoretically shown in Section III-C, we now experimentally show estimation of physical properties from pouring sound.

a) *Evaluation on air column length:* We evaluate our pitch detection network to estimate the air column length. We compare against simpler baselines for pitch detection. We try classical pitch detectors like Yin [95], CNN-based detectors like CREPE [91] and recent self-supervised detectors like PESTO [96]. We also try a baseline with per-frame argmax on the spectrogram. Note that other baselines estimate pitch directly from raw audio samples while the argmax method operates on a spectrogram. Given pitch (fundamental wavelengths) at certain

TABLE III

CO-SUPERVISION IMPROVES PHYSICAL PROPERTY ESTIMATION. MEAN ABSOLUTE ERROR IN ESTIMATING VARIOUS PHYSICAL PROPERTIES. OUR VISUALLY CO-SUPERVISED MODEL GENERALLY IMPROVES OVER THE SYNTHETIC-TRAINED MODEL IN ESTIMATING PHYSICAL PROPERTIES FROM PITCH. WE OBSERVE NOTICEABLE IMPROVEMENTS IN ESTIMATING RADIUS AND FLOW RATE. THIS SUGGESTS CO-SUPERVISION PARTICULARLY IMPROVES ESTIMATION OF PITCH TOWARDS THE END OF THE AUDIO AS WELL AS THE SLOPE OF PITCH GENERALLY.

Property	Unit	Not.	Test set I		Test set II	
			Syn.↓	Co-sup.↓	Syn.↓	Co-sup.↓
Static properties						
Height	cm	H	2.23	2.27	2.77	2.85
Radius	cm	R	1.62	1.39	2.24	1.88
Dynamic properties						
Flow rate	ml/s	Q	25.20	22.50	45.70	40.41
	s	$\tau_{\frac{1}{4}}$	3.96	4.16	4.39	4.10
TTF	s	$\tau_{\frac{1}{2}}$	1.62	1.49	3.44	2.99
	s	$\tau_{\frac{3}{4}}$	1.53	1.07	2.66	2.21

time points, we fit a line to estimate $\lambda(t)$ on the collection of obtained points using RANSAC to avoid outliers.

To compute length of air column from wavelengths, we rely on (5). We compare our models with the baselines in estimating $l(t)$ and report the mean absolute error averaged over all time points. The results are reported in Table II. Our models outperform all the baselines. Moreover, the co-supervised model achieves an error of 0.60 cm as compared to the audio-only variant's 0.78 cm on Test set I. The results are similar on the more challenging Test set II.

b) *Evaluation on other physical properties:* Having shown the superiority of our pitch detection model, we now check how much visual co-supervision helps in estimating other physical properties in Table III.

- *Container dimensions:* We obtain the container radius R and height H using (6). The ground truths are obtained by manually measuring the said containers. Our best model achieves an MAE of 2.27 cm in height and 1.39 cm in radius on Test set I and 2.77 cm / 1.88 cm on Test set II. Note that radius measurement depends on reliable wavelength estimates towards the end of the audio which is significantly better in case of a co-supervised model than the audio-only variant. This is further shown qualitatively and quantitatively in Fig. 8.
- *Volume flow rate:* Following (7), we evaluate volume flow rate prediction. This is dependent on R and derivative of $\lambda(t)$. The ground truth is obtained using the ratio of the volume of container and the time it took to fill it completely. The co-supervised model achieves 22.5 ml/s compared to 25.2 ml/s of the audio-only model on Test set I. Similar improvement is observed on Test set II.
- *Time to fill:* Following (8), we evaluate on time to fill prediction. We are given a partial audio, i.e., cut at 25% or 50% or 75% of its original duration and the task is to predict time (in seconds) that it will require to fill the container. Since we vary the flow rates randomly, it prevents the obvious shortcut of simply extrapolating based on the given

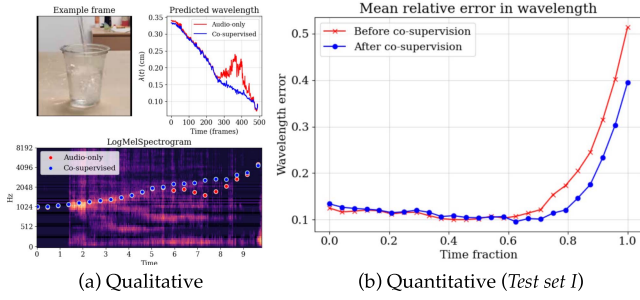


Fig. 8. Where does co-supervision help the most? We find that co-supervision is most beneficial towards the end of the audio where the signal is generally low. This is shown qualitatively on a sample in (a) and quantitatively on Test set I in (b). This helps in more precise radius estimation since it depends on $\lambda(T)$. Moreover, qualitatively, and quantitatively in estimating flow rate, it seems to help with more reasonable slopes of λ .

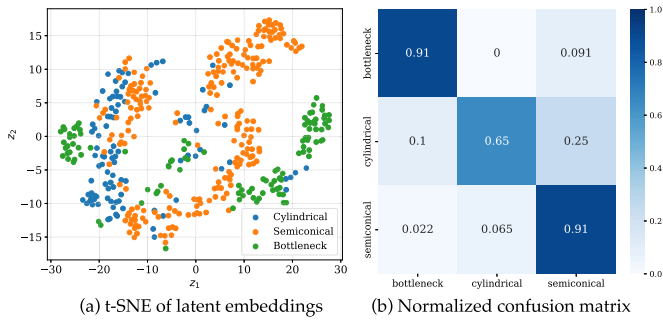


Fig. 9. Results on shape recognition from audio embeddings. (a) Latent embeddings learned by the co-supervised audio model encode container shape. (b) On an unseen test set, it is able to recognize the container shapes with a sample accuracy of 90.91% and mean class accuracy of 92.47%. Some semiconical containers with small difference between the top and base radius are essentially cylindrical and that shows in the t-SNE embeddings.

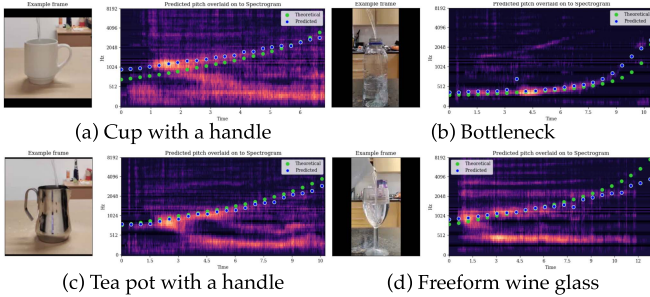


Fig. 10. Generalization across container shapes. Although our pitch predictor is trained only on cylinder-like containers, it works reasonably well on various free-form shapes encountered in daily use (unseen during training). The theoretical estimate (green curve) is obtained assuming a cylinder or a bottle and is an imperfect measure of the pitch. Nonetheless, the prediction (blue curve) tracks the pitch accurately even though it may disagree with the theoretical estimate.

time. As shown in Table III, the co-supervised model is comparable to the audio-only when provided with only 25% of the input but outperforms the latter given 50% or 75% of the input on Test set I. On Test set II, the co-supervised model outperforms at all levels of the input.

We also conduct an experiment to evaluate the impact of height at which liquid is being poured. Since pouring height only

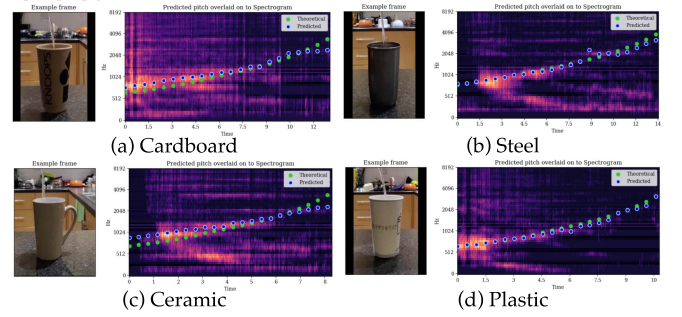


Fig. 11. Generalization across container materials. Our pitch detector works well for diverse kinds of container materials while the shape (semi-conical) is fixed. The theoretical estimate (green curve) is obtained by assuming it as a cylinder and thus is not perfect and only used for reference.

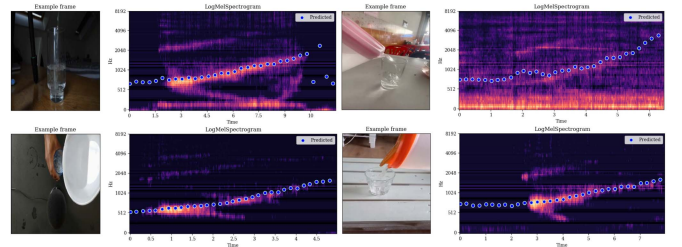


Fig. 12. Generalization to in-the-wild videos. We qualitatively evaluate on YouTube videos. Our pitch detector works very well even on these samples. Notice the variability in the visual inputs and contrast that to the consistency in the audio recordings.

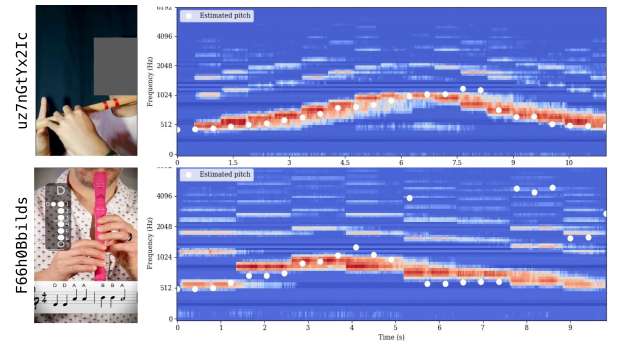


Fig. 13. Generalization to music. Qualitatively, we find surprisingly reasonable generalization to flute sounds likely since the same underlying physical phenomenon produces resonance. We do not find generalization to other musical instruments or polyphonic music due to a large domain gap. The YouTube IDs of the samples are provided.

affects magnitude, it should not affect estimation of physical properties. We demonstrate this in practice by showing a low correlation ($\rho = -0.2$) between pouring height and error in container height estimation. More details are in the Appendix, available online.

B. Recognizing Liquid Container Shape From Its Pouring Sound

Differently shaped containers exhibit different structures of the resonance curves. Sound of pouring in cylindrical containers

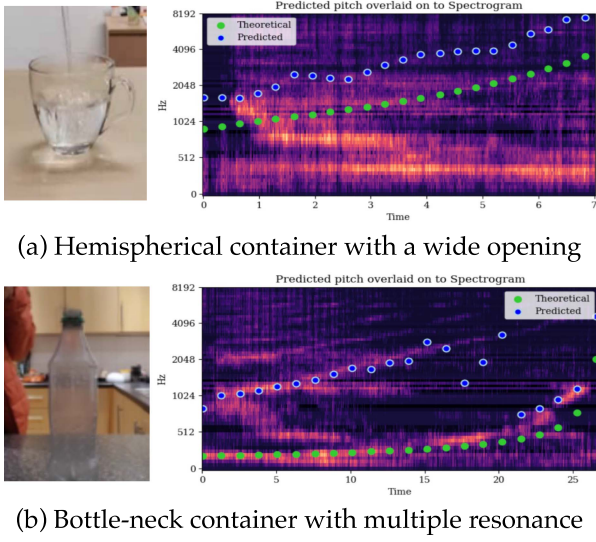


Fig. 14. Failure cases. (a) The model struggles in cases of hemispherical containers that have too much room for air to pass leading to very weak resonance (b) another challenging case is that of bottle-neck containers that show more than one mode in the frequency distribution on spectrogram.

show a pitch that varies as $1/l(t)$ where $l(t)$ is the length of air column [1]. In contrast, in bottles, pitch varies as $1/\sqrt{l(t)}$ [10]. Though we train for pitch detection only on cylinder-like containers, here we evaluate whether the learned representations encode container shape. Formally, given the sequence of features of the sound of pouring, the task is to classify whether the shape of the container is cylindrical, semiconical, or bottleneck.

Given the sound of pouring, we first compute features from the co-supervised audio encoder. These are outputs, say $\{z_i\}_{i=1}^N$, from the last Transformer block of `wav2vec2` before going into the prediction head in Fig. 3. Then, we concatenate the following vectors to create a single summary vector per sample: the mean of the sequence and each of the vectors at 25%, 50% and 75% of the sequence length.

$$\mathbf{z}_{\text{summary}} = \text{concat} \left(\mathbb{E}_t[\mathbf{z}_t], \mathbf{z}_{\frac{N}{4}}, \mathbf{z}_{\frac{N}{2}}, \mathbf{z}_{\frac{3N}{4}} \right), \quad (12)$$

where N is the sequence length. We attach a 3-way linear shape classifier head to this vector.

To report performance, we consider the *Test Set III* set of our dataset consisting of 434 videos (227 semiconical, 107 bottleneck and 100 cylindrical). These are unseen samples not part of the training and evaluation set used previously to estimate physical properties. We split these randomly in an 80-20 split. On this split, we achieve a per sample accuracy of 90.91% and a mean class accuracy of 92.47%. t-SNE [97] embeddings and the normalized confusion matrix are shown in Fig. 9. In comparison, features without co-supervision achieve 88.63% and 89.44%. This shows: (i) a model trained to detect pitch implicitly encodes container shape, and (ii) co-supervision further improves shape recognition.

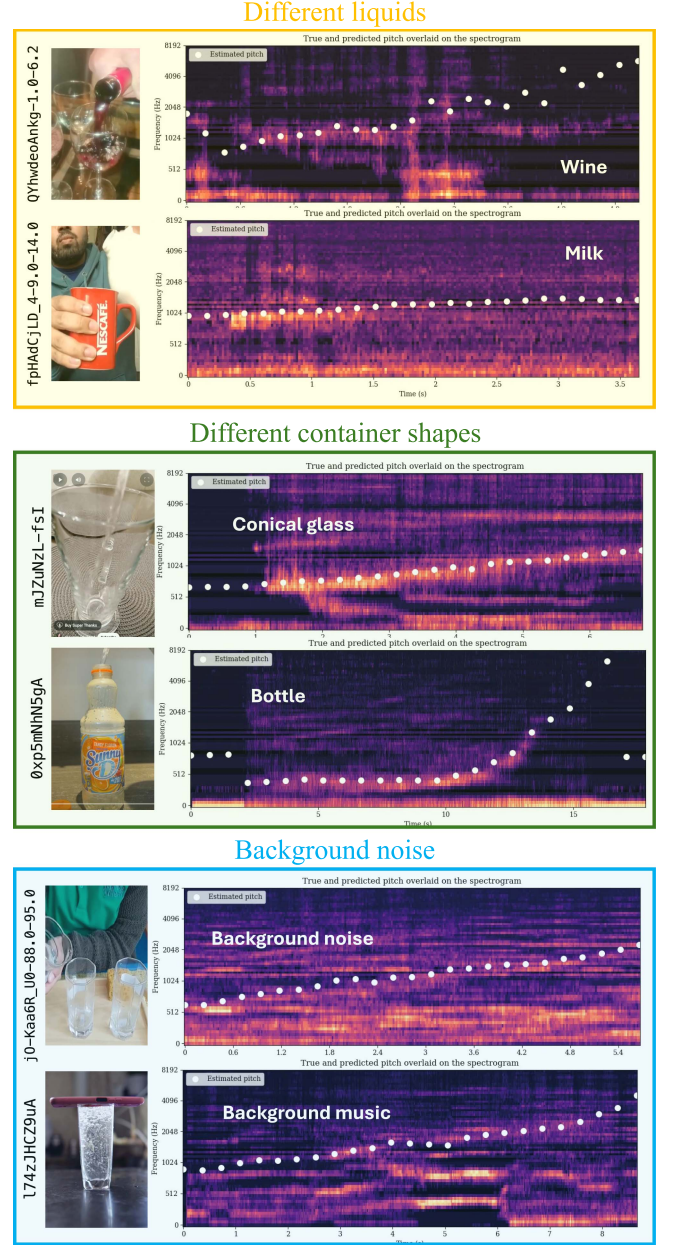


TABLE IV

ESTIMATING LIQUID WEIGHT FROM ITS POURING SOUND. AVERAGE MAE IN MASS ESTIMATION (IN OUNCES, (OZ)) FROM SHORT SNIPPETS OF THE SOUND OF POURING ACROSS DIFFERENT CONTAINER AND LIQUID CONFIGURATIONS FROM THE DATASET BY WILSON ET AL. [46]. LINEAR PROBING OUR PRE-TRAINED CO-SUPERVISED FEATURES OUTPERFORMS ALL THE BASELINES AND PERFORMS COMPETITIVELY TO THE STRONG SUPERVISED FINE-TUNING BASELINE IN [46]. NOTE THAT VISUAL INFORMATION IS NOT USED IN ANY FORM. WE PRE-TRAIN ON OUR DATASET AND LINEAR PROBE ON THE EVALUATION DATASET WHILE WILSON ET AL. [46] FINE-TUNE ON THIS DATASET. BEST VIEWED ON A SCREEN WITH ZOOMING IN.

Material	Plastic	Glass	Porcelain	Metal	Glass	Glass	
Shape	Semicone	Semicone	Pyramid	Cylinder	Bottle	Bottle	
Liquid	Water	Water	Water	Water	Milk	Water	
Sample image							
							Avg
k-NN	3.4	3.6	2.2	2.5	2.7	2.7	2.85
Linear SVM	3.4	4.8	3.3	4.1	3.5	4.3	3.90
SoundNet-5 [71]	3.4	4.2	4.4	4.7	3.0	3.6	3.88
SoundNet-8 [71]	3.2	6.1	3.5	4.2	5.8	4.7	4.58
TCN [98]	1.5	1.9	2.0	1.7	3.9	3.7	2.45
PSNN [46]	1.2	1.2	1.3	0.7	1.8	1.9	1.35
Ours	<u>1.3</u>	<u>1.3</u>	<u>1.4</u>	<u>0.9</u>	1.0	1.3	1.20

of features $\{\mathbf{z}_i\}_{i=1}^N$. We attach a linear regressor head which outputs a scalar weight for each feature vector $\mathbf{z}_i$.

For transparent comparison, we report comparison with the same baselines, on the same data and splits as in [46]. Specifically, it consists of 136 video sequences with weight annotations across six different containers and two liquids. Following [46], for each container, we train a separate regressor on its own split and report performance in Table IV. Our model outperforms all the baselines as well as the strong supervised baseline in [46]. On average, our model achieves an MAE of 1.20 oz which is the best amongst all baselines. Not surprisingly, our model performs the best with a cylindrical container (Column 4) but also is impressive on a bottleneck container (Columns 5 and 6). This shows that a network trained for pitch detection learns features capable of estimating liquid mass which is a function of the container volume and liquid density. In the Supplemental, we also evaluate generalization across containers where we train a mass regressor on sounds of one container and evaluate it on those of another container.

D. A Case Study on Generalizability

While our pitch detection model is trained on cylinder-like transparent containers of glass and plastic in a clean setting, we study in detail its ability to generalize to previously unseen scenarios: e.g., different container shapes, container materials, and to samples from other datasets (e.g., [46]). We even probe its ability to detect pitch in unseen domains such as music even though it is solely trained on the sounds of pouring.

To test generalization across shapes, we evaluate on various (unseen) containers of different shapes from our dataset (e.g., a cup or a teapot) and report robust results in Fig. 10. To test generalization across materials, we pick containers of the same shape (semi-conical) of various materials (e.g., steel, ceramic,

cardboard, etc.) from our dataset. We find fairly convincing evidence for generalization across materials in Fig. 11. Finally, we also evaluate on samples in-the-wild from YouTube. As shown in Fig. 12, the predictions fairly accurately track the fundamental frequency curve. Note the variability in the video which makes it hard to adjudge the container shape or size from vision, but it is much easier to infer from the detected pitch from our model. More examples of robustness to different liquids, container shapes and background noise is shown in Fig. 15.

a) *Generalization to music*: While pitch detectors trained on music data do not generalize on pouring sounds (Table II), it worth asking if the reverse is true. Given the large domain gap, we do not expect the model to work in an entirely different domain (like polyphonic music). However, as shown in Fig. 13, it is slightly surprising that the pitch detector trained only on pouring sounds does reasonably well on flute sounds. This is likely because the same underlying principle that produces sound in both cases. It does not generalize to other musical instruments or polyphonic music.

b) *Failure cases*: Our pitch detector model struggles with hemispherical containers (e.g., a bowl as shown in Fig. 14(b)). In such cases, there is a lot of room for air to escape which prevents the sort of “filling-up” effect seen in other containers. It also struggles with certain cases where multiple frequency modes are present. For example, in some bottleneck containers (Fig. 14(a)), along with the fundamental, a strong linear frequency curve is also present. The model fails to pick both together.

VII. CONCLUSIONS AND EXTENSIONS

In this work, we considered the case of pouring water in a container and analysed the connection between the underlying physics and audio-visual observations. While humans can reason well about the physical properties (e.g., absolute container size) merely from the sound of pouring, we have shown early evidence of training machines to achieve similar capabilities. We developed synthetic data to pre-train an audio network (Wav2Vec2.0) for pitch detection. We fine-tuned it on real data with no external supervision thanks to the co-supervision from the video stream. We demonstrated that basic physical properties can be recovered accurately from the estimated pitch. We showed that the co-supervised representations also encode other useful properties such as container shape and liquid weight. Finally, we showed promising generalization to different container shapes, materials, and in-the-wild videos from YouTube, e.g., radial resonance as described in Section III-B [99], [100]. We hope that our work also prompts similar studies for physical understanding from the sound of other mundane activities.

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